

Midterm 2

ORF 309, Spring 2025

Instructions: This midterm has 3 questions for a total of 90 points. Partial credit will be awarded for partial proofs or arguments. Strive to reduce answers as much as possible (e.g., give a fraction in simplest terms, though a numerical answer requiring a calculator is not necessary). You may freely use basic facts from class, such as laws of probability, properties of conditional expectation, or manipulations with pmfs/pdfs. However, you must provide adequate justifications for important steps. Put another way, try your best to demonstrate awareness that you are applying the correct facts in the correct way by explicitly mentioning the results you are applying, indicating the objects they are applied to, and checking the required assumptions.

Length: 105 minutes (for working on the problems) + 15 minutes (for submission)

Good Luck!

Question	Points	Score
1	30	
2	30	
3	30	
Total:	90	

1. Alice is planning a trip to California and is observing the price movement of flights that depart from Newark Liberty International Airport (she is quite flexible about travel dates). If the price goes down to 200 dollars, she will purchase the flight ticket and start packing her belongings. However, if the price goes up to 900 dollars, it will be so expensive that she will give up and cancel the travel plan. For each hour, the price either increases by 2 dollars with probability $\frac{1}{3}$ or decreases by 1 dollar with probability $\frac{2}{3}$. We assume the price movements within different hours are mutually independent.
 - (a) (10 points) Assume the current price is 500 dollars. What is the expected price after 24 hours?
 - (b) (10 points) Let $p(i)$ denote the probability that Alice gives up traveling before purchasing the ticket, given the current price is i dollars. Use first-step analysis to rigorously derive a recursive formula for $p(i)$, $200 < i < 900$.
 - (c) (10 points) Let $\mu(i)$ denote the expected number of hours until Alice makes any decision (either she purchases the ticket or gives up traveling). Use first-step analysis to rigorously derive a recursive formula for $\mu(i)$, $200 < i < 900$.

2. Flights at Newark Liberty International Airport often experience unpredictable delays and, according to the Federal Aviation Administration, the largest cause of air traffic delays in the National Airspace System is weather. Let the weather condition be quantitatively modeled as a single index (called the weather index) Z , and a higher value of Z means better weather. Suppose the weather index Z is a random variable that is exponentially distributed with mean 1. Given this index Z , the flight delay time T (in hours) will be exponentially distributed with mean $1/Z$. In other words,

$$\mathbb{P}[T > t \mid Z = z] = e^{-zt}, \quad \forall t, z > 0.$$

- (a) (10 points) Determine the joint probability density function for the pair of random variables (T, Z) .
- (b) (10 points) Suppose the weather condition is better than average (that is, $Z > 1$). Given this information, what is the probability that the flight will be delayed by more than 2 hours?
- (c) (10 points) Suppose the current flight is delayed by exactly 2 hours. Given this information, what is the probability that the weather condition is better than average (that is, $Z > 1$)?

3. Many travelers who have landed at Newark Liberty International Airport will take taxis to get to their destinations. At the taxi stop, passengers arrive according to a Poisson process with a rate 2 passengers per minute (they join the queue and wait until a taxi is ready for them), and taxis arrive according to a Poisson process with a rate 1 vehicle per minute (they pick up one passenger each and wait at the stop if there are no waiting passengers). We assume that the arrivals of travelers and taxis are independent and that currently there are no travelers or taxis waiting at the stop.
- (a) (10 points) What is the probability that at least 10 passengers arrive before the first taxi comes?
 - (b) (10 points) What is the expected time until the first taxi arrives, given that by the time it arrives there are exactly 10 passengers waiting at the stop?
 - (c) (10 points) Accidents, which lead to temporary traffic breakdowns, occur, independently of the arrivals of travelers and vehicles, according to a Poisson process with a rate of once per 100 minutes. What is the probability that at least one passenger is picked up by a taxi before any accident occurs?